From Noise to Diversity: Random Embedding Injection in LLM Reasoning

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Abstract

Recent *soft prompt* methods, which prepend or insert trainable continuous embeddings into LLM inputs as virtual tokens, optimize these embeddings to enhance mathematical reasoning. However, the mechanism by which embedding injection itself affects model behavior, independent of optimization, remains unexplored. We investigate Random Soft Prompts (RSPs), continuous vectors sampled from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table and injected without any training. Empirically, RSPs increase early-stage token entropy while generation stabilizes thereafter, and combining RSPs with temperature sampling yields higher trajectory diversity as measured by Pass@n. Theoretically, we show that the RSP contribution at each attention layer is exactly captured by a scalar random-token attention mass that is structurally diluted as autoregressive cache growth accumulates real tokens, and that under a local affine analysis of the perturbed logits RSPs can open previously excluded tokens into the effective top-K candidate set in a way temperature sampling provably cannot. The connection to a Hopfield-style energy landscape is recorded in the appendix as an interpretive lens but is not used in any formal result. We further discuss the implications for GRPO training, where trajectory diversity is critical for learning signal quality.

1 Introduction

As large language models (LLMs) scale to billions of parameters, full fine-tuning becomes increasingly expensive. This has motivated parameter-efficient methods that adapt frozen models by introducing a small number of trainable parameters [Hu et al., 2022, Houlsby et al., 2019]. Among these, *soft prompts* prepend or insert learnable continuous embeddings into the input sequence, steering model behavior through the attention mechanism [Li and Liang, 2021, Lester et al., 2021]. This paradigm has been actively adopted for enhancing mathematical reasoning in LLMs [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2026]. Despite their diverse designs, these methods share a common structure: they inject continuous embeddings that lie outside the pretrained vocabulary and optimize them to improve downstream performance.

However, we find that optimization is not a necessary condition for these effects. We observe that injecting randomly initialized, untrained continuous embeddings can also shift the output distribution. For instance, applying a chat template to a base model not fine-tuned with such format causes a prompt-format mismatch that degrades reasoning capability, but appending random embeddings to such mismatched inputs mitigates this degradation. This suggests that injecting out-of-vocabulary continuous embeddings itself, independent of any learned content, plays a non-trivial role in shaping model behavior. Existing studies primarily evaluate *soft prompt* methods through end-task accuracy, and the mechanisms by which embedding injection influences the output distribution remain insufficiently explored.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

In this work, we investigate the role of Random Soft Prompts (RSPs), continuous vectors sampled from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table and injected without any training. Empirically, RSPs increase early-stage token entropy while the output stabilizes in later generation stages, and combining RSPs with temperature sampling yields more diverse reasoning trajectories as measured by Pass@n scaling. Theoretically, we identify an attention-level mechanism in which the RSP contribution is exactly captured by a scalar random-token attention mass that is structurally diluted by autoregressive cache growth, and we show that under a local affine analysis of the perturbed logits RSPs can open previously excluded tokens into the effective top-K candidate set in a way temperature sampling cannot. We further discuss the implications of these findings for Group Relative Policy Optimization (GRPO) [Shao et al., 2024], where trajectory diversity is essential for effective learning signals.

Our contributions are as follows:

- We empirically analyze how RSPs affect LLM generation, showing that RSPs increase early token entropy and produce more diverse reasoning trajectories when combined with temperature sampling.
- We give a transformer-level mechanism in which the RSP contribution is controlled by a scalar attention mass with structural cache-driven dilution, and a distributional analysis under a local affine surrogate showing that RSP can open previously excluded tokens into the effective top-K candidate set in a way temperature sampling cannot.
- We discuss the implications of RSP-induced diversity for GRPO training and argue that RSP provides exploration gains complementary to existing temperature-based approaches.

2 Background

2.1 Soft Prompt Methods for LLM Reasoning

Soft prompts emerged as a parameter-efficient alternative to full fine-tuning. Prefix-Tuning [Li and Liang, 2021] introduced the paradigm by prepending trainable continuous vectors to every transformer layer, Prompt Tuning [Lester et al., 2021] simplified this to input-layer-only tuning and showed that it matches full fine-tuning at sufficient scale, and P-tuning [Liu et al., 2024] extended the approach with an LSTM-based prompt encoder to model inter-position dependencies.

More recently, *soft prompts* have been actively applied to LLM reasoning. TTSV [Kang et al., 2026] optimizes prefix embeddings at test time via entropy minimization, steering the model toward high-certainty states to activate latent reasoning capabilities. SoftCoT [Xu et al., 2025] replaces explicit chain-of-thought tokens with soft tokens generated by an assistant model, leveraging the stronger representational capacity of continuous representations over discrete language. LTPO [Ye et al., 2026] optimizes latent thought embeddings per test instance through policy gradient with a confidence-based intrinsic reward. COCONUT [Hao et al., 2025] feeds the model's own hidden states back as input embeddings, enabling reasoning in a continuous latent space. MemGen [Zhang et al., 2026] demonstrates that embedding vectors can serve as experience memory by constructing latent token sequences that enrich the model's reasoning. Pause tokens [Goyal et al., 2024] insert learnable tokens into the input to provide additional computation steps.

These approaches all inject out-of-vocabulary continuous embeddings optimized for task-specific objectives. Because evaluation centers on end-task accuracy, the contribution of the injection itself and that of the optimization are not separated, and the mechanisms by which such out-of-vocabulary continuous embeddings shape model behavior remain a separate and underexplored question.

2.2 Noise Injection in Neural Networks

Neural networks have incorporated noise at various stages. During training, dropout [Srivastava et al., 2014] randomly deactivates hidden units to prevent overfitting, and NEFTune [Jain et al., 2024] adds noise to token embeddings during instruction fine-tuning, reporting improvements in instruction following. These methods use noise as a regularizer and do not apply perturbations at inference time.

At inference time, perturbation-based approaches primarily target robustness and safety. Randomized smoothing [Cohen et al., 2019] injects Gaussian noise into inputs to obtain certified robustness

Table 1: Accuracy (%) across model configurations and benchmarks. RSP reports the best result across injection positions (Prefix, Infix, Suffix). Full per-position breakdown is in Appendix A. Values in parentheses indicate the difference from the baseline.

	MATH-500		GSM8K		AIME24	
Model	Baseline	RSP	Baseline	RSP	Baseline	RSP
Instruct Models						
LLaMA-3.1-8B-Inst	52.20	49.40 (−2.8)	85.75	86.73 (+1.0)	6.67	10.00 (+3.3)
Qwen2.5-Math-7B-Inst	83.20	84.20 (+1.0)	95.45	95.68 (+0.2)	13.33	16.67 (+3.3)
Qwen2.5-Math-1.5B-Inst	73.00	75.20 (+2.2)	85.29	85.75 (+0.5)	10.00	20.00 (+10.0)
Base Models (Raw Text)						
Qwen2.5-Math-7B	72.20	71.60 (−0.6)	84.15	84.31 (+0.2)	6.67	16.67 (+10.0)
Qwen2.5-Math-1.5B	61.40	64.40 (+3.0)	77.86	74.30 (−3.6)	16.67	10.00 (−6.7)
Base Models + ChatML (Format Mismatch)						
Qwen2.5-Math-7B	52.20	70.40 (+18.2)	58.30	75.97 (+17.7)	23.33	23.33 (+0.0)
Qwen2.5-Math-1.5B	34.40	63.40 (+29.0)	37.45	67.02 (+29.6)	13.33	20.00 (+6.7)

guarantees. In the LLM setting, SmoothLLM [Robey et al., 2025] perturbs input prompts at the character level to defend against adversarial jailbreaking. These methods require multiple noisy forward passes with output aggregation, and target prediction stability rather than output diversity.

In reinforcement learning, NoisyNet [Fortunato et al., 2018] injects learnable noise into network weights to improve exploration, demonstrating that the location and structure of noise injection affect exploration quality. This motivation is closest to RSPs, which similarly target diversity, but differ in injection target (input embeddings vs. network weights) and require no learned noise parameters.

RSPs differ from these approaches in the following respects. **(1)** RSPs operate solely at inference without any training, whereas dropout and NEFTune inject noise during training, modifying the resulting model parameters accordingly. **(2)** RSPs preserve the original input and append fresh embedding vectors in a single forward pass, whereas robustness methods corrupt the input and aggregate over multiple noisy passes. **(3)** RSPs use purely random vectors sampled from the pretrained embedding statistics, whereas NoisyNet learns noise parameters through optimization.

3 Experimental Analysis

3.1 Random Soft Prompts: Definition and Experimental Setup

Let $\mathbf{W}_E \in \mathbb{R}^{V\times d}$ denote the token embedding matrix of a pretrained LLM, where V is the vocabulary size and d is the hidden dimension. We define an RSP $\mathbf{H} = [h_1, \ldots, h_L] \in \mathbb{R}^{L\times d}$ as a sequence of L continuous vectors, where each vector is independently sampled from an isotropic Gaussian distribution whose mean and variance are matched to those of the pretrained token embedding matrix:

$$h_j \sim \mathcal{N}(\mu_E,\ \sigma_E^2 \mathbf{I}), \quad j = 1, \ldots, L, \tag{1}$$

with $\mu_E = \frac{1}{Vd}\sum_{i,k}[\mathbf{W}_E]_{i,k}$ and $\sigma_E = \mathrm{std}(\mathbf{W}_E)$. Let $\mathbf{X} = [x_1, \ldots, x_T] \in \mathbb{R}^{T\times d}$ denote the embedding sequence of the original input. The RSP-augmented input $\tilde{\mathbf{X}}$ is constructed by concatenating $\mathbf{H}$ with $\mathbf{X}$ at one of three positions without modifying the original prompt text: *prefix* ($[\mathbf{H}; \mathbf{X}]$), *infix* ($\mathbf{H}$ inserted within $\mathbf{X}$), or *suffix* ($[\mathbf{X}; \mathbf{H}]$), following the injection strategies adopted in prior work [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Zhang et al., 2026]. Sampling a new $\mathbf{H}$ for each batch ensures that the results are not dependent on any realization of the random embeddings.

We evaluate on mathematical reasoning benchmarks: MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024] and the Qwen2.5-Math model family [Yang et al., 2024] across instruct models, base models, and base models with mismatched chat templates. All experiments use greedy decoding to isolate the effect of RSPs from sampling stochasticity. Full experimental details are provided in Appendix A.

Table 2: Comparison with existing soft prompt methods (TTSV, SoftCoT, LTPO) under unified evaluation. Baseline and RSP values are from Table 1. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Benchmark	Baseline	RSP	TTSV	SoftCoT	LTPO
Qwen2.5-Math-7B-Instruct	MATH-500	83.20	**84.20**	83.80	82.80	83.00
	GSM8K	95.45	**95.68**	95.30	95.00	94.84
	AIME24	13.33	16.67	**23.30**	16.67	13.33
Qwen2.5-Math-1.5B-Instruct	MATH-500	73.00	75.20	75.20	**76.00**	72.40
	GSM8K	85.29	**85.75**	85.70	84.46	84.53
	AIME24	10.00	**20.00**	6.70	6.67	10.00

3.2 How Does Untrained RSP Compare to Baselines & Optimized Soft Prompts?

We first examine how injecting random embeddings affects the model's reasoning performance. Table 1 reports the accuracy for each model configuration, using the best RSP result across injection positions (Prefix, Infix, Suffix) and $L \in \{10, 15, 20\}$ for each benchmark.

For instruct models and base models with their native prompt formats, RSP injection maintains or slightly shifts baseline performance within a few percentage points. Most strikingly, under prompt-format mismatch — where base models receive ChatML templates they were not trained on — RSP yields pronounced improvements of up to $+29$ pp. These gains are difficult to attribute to simple perturbation effects, as noise that merely raises model uncertainty would further degrade performance.

We compare RSPs with prior soft prompt methods (TTSV, SoftCoT, LTPO) that optimize their embeddings for distinct objectives, under a unified evaluation protocol reported in Table 2. All methods are re-evaluated with the same unified extraction pipeline (Appendix A.2), enabling a format-agnostic comparison despite each method's distinct prompt format and grading scheme. The results place RSPs on par with the optimized methods without requiring any training.

3.3 How Does RSP Affect the Output Distribution?

We analyze how RSP injection changes the model's output distribution on Qwen2.5-Math-1.5B-Instruct with MATH-500, measuring per-token entropy, top-1 probability, and varentropy across the generation process. Figure 1 compares these metrics for the first 5% of generation steps across RSP variants and prior soft prompt methods (LTPO, SoftCoT, TTSV). Full statistics are in Appendix A.4.

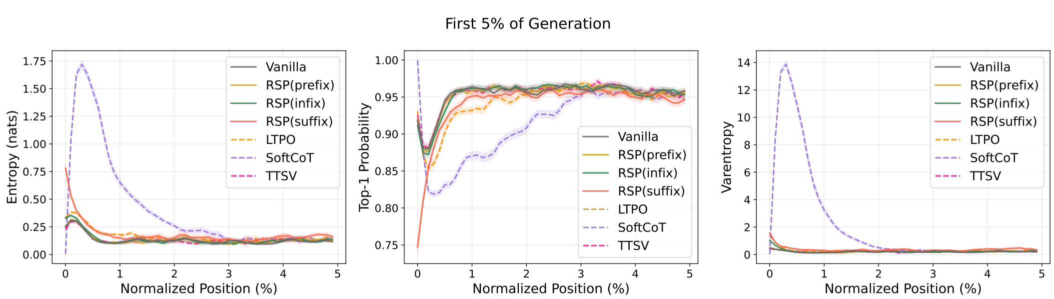

Figure 1: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: prior soft prompt methods (LTPO, SoftCoT, TTSV).

These results show that RSP modifies the early-stage output distribution across all injection positions: entropy increases (the token distribution flattens), top-1 probability decreases (commitment to a dominant token weakens), and varentropy increases (indicating multimodal distributions [Ahmed et al., 2026]). High-entropy positions are known to act as critical forking points that determine reasoning trajectories [Wang et al., 2025], so this shift moves the model toward a state where multiple reasoning paths can coexist. The change is localized to the initial reasoning stage. In the middle and final portions of generation, all three metrics converge to baseline levels.

Although existing soft prompt methods are optimized for different objectives, they share a common pattern of increasing early-stage entropy. Notably, TTSV, which optimizes soft prompts using a reward that reduces the mean entropy of reasoning trajectories, still exhibits early entropy comparable to the baseline. This suggests that elevating early-stage entropy is an inherent effect of injecting non-pretrained continuous embeddings, independent of the optimization objective.

3.4 Does RSP Induce Trajectory Diversity?

Section 3.3 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. The three analyses converge on a directional picture: RSP's first-token perturbation propagates into semantically more diverse reasoning trajectories, which in turn yields greater output diversity at the task level. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, suffix injection, and $L = 10$.

Token-level: distribution beyond temperature. We test whether RSP modifies the first-token distribution in a qualitatively different way from temperature scaling, which preserves token rankings while only flattening probabilities. Under autoregressive decoding, any trajectory-level divergence between RSP and the baseline must originate from the first generation step, a critical forking point for reasoning trajectories [Wang et al., 2025]. Using the baseline at τ=1.0 as reference, we compare RSP at τ=1.0 against the baseline at τ=2.0 (the high-temperature ceiling of what flattening alone can achieve) on three complementary metrics: Spearman rank correlation ρ to detect rank reorganization, the probability mass placed outside the baseline's top-10 to measure support expansion, and Jensen–Shannon divergence to quantify the overall magnitude of distributional change (formal definitions in Appendix B).

Table 3: First-token distribution metrics measured against the baseline at τ=1.0, comparing the baseline at τ=2.0 with RSP at τ=1.0.

Metric	τ=2.0	RSP
Spearman ρ	1.0	**0.651**
Mass outside top-10	5.15%	**27.64%**
JS divergence	0.086	**0.231**

Table 3 reports the contrast. Even at τ=2.0, only 5.15% of probability mass falls outside the baseline's top-10. RSP at τ=1.0 reaches 27.64%. The Spearman rank correlation tells the story: because temperature scaling $p_i^{(\tau)} \propto p_i^{1/\tau}$ is a monotone transform that preserves token ranking ($p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$) for any $\tau > 0$, ρ relative to the baseline is exactly 1. RSP, by contrast, reranks the token distribution itself, reducing ρ to 0.651—an effect temperature cannot produce at any τ. The Jensen–Shannon divergence (0.231) quantifies the overall magnitude of this distributional change. Since $\rho < 1$ rules out rank-preserving flattening as the sole cause, the high divergence reflects a structural reshape of the distribution rather than a uniform flattening of the same ranking.

Trajectory-level: semantic diversity. Token-level reranking raises a second question: do first-token perturbations produce semantically diverse reasoning trajectories, or do independently perturbed trajectories converge to semantically similar ones? We randomly sample 64 problems from MATH-500 and generate 300 independent trajectories per problem under Baseline and RSP at temperature $\tau = 1.0$. Each trajectory is encoded into a dense semantic vector by BGE-M3 [Chen et al., 2024], a sentence-level multilingual embedding model, and we compute three complementary metrics over these embeddings: *pairwise cosine distance* between trajectories captures the overall semantic spread of the trajectory set; *inter-cluster distance* between centroids identified by DBSCAN [Ester et al., 1996], a density-based clustering algorithm that groups nearby trajectories without prespecifying the number of clusters, captures the separation between distinct trajectory groups; and *intra-cluster distance* captures semantic coherence within each group.

Table 4: Semantic diversity metrics (64 problems, 300 trajectories per condition).

Metric	Baseline	RSP
Pairwise cos. dist.	0.0612	**0.0879**
Inter-cluster dist.	0.0183	**0.0982**
Intra-cluster dist.	0.0526	0.0528

Table 4 reveals three concurrent patterns. Pairwise cosine distance rises by about $1.4\times$, indicating that the trajectory set as a whole becomes more semantically diverse. Inter-cluster distance jumps by about $5.4\times$, showing that the trajectories split into substantially more separated groups. Intra-cluster distance is essentially unchanged, indicating that semantic coherence within each group remains comparable to the baseline. RSP also yields larger pairwise distance than baseline on 56 of 64 problems (87.5%).

Outcome-level: Pass@n scaling. Finally, we evaluate whether the token- and trajectory-level diversity translates into task-level outcome diversity through Pass@n [Chen et al., 2021], the probability that at least one of n sampled solutions is correct. We compare four conditions with $n = 16$ samples per problem on MATH-500 and AIME24: *(i) Baseline,* $\tau = 1.0$: temperature sampling only; *(ii) RSP (fixed seed),* $\tau = 1.0$: a single RSP shared across all samples; *(iii) RSP (indep. seed), greedy*: a different RSP per sample without temperature; and *(iv) RSP (indep. seed),* $\tau = 1.0$: a different RSP per sample combined with temperature sampling.

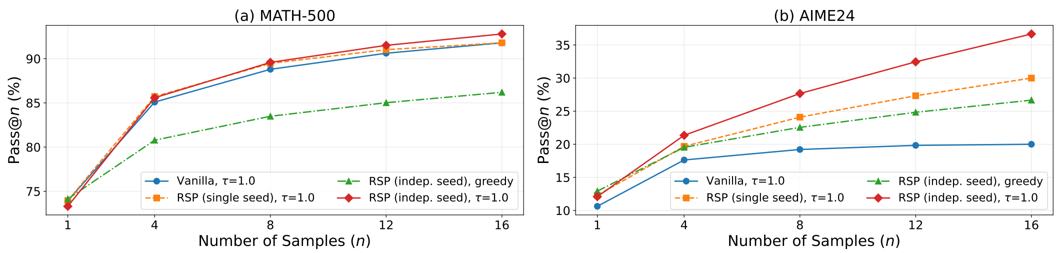

Figure 2: Pass@n scaling on (a) MATH-500 and (b) AIME24 with Qwen2.5-Math-1.5B-Instruct, $n = 16$ samples per problem. Baseline: temperature sampling only. Fixed seed: single RSP shared across samples combined with temperature. Indep. seed: a different RSP per sample, with or without temperature.

Condition *(iv)* achieves the highest Pass@n on both benchmarks, reaching 92.80% on MATH-500 and 36.67% on AIME24 at $n = 16$. Condition *(ii)* shows no improvement over the baseline, suggesting that the diversity gain likely depends on varying the random embeddings across samples rather than a single fixed perturbation. Pass@1 remains comparable to the baseline on MATH-500, and on the more challenging AIME24, *(iv)* achieves higher Pass@1 than the baseline. Together these observations suggest that combining independent RSPs with temperature sampling may yield greater trajectory diversity while preserving single-sample accuracy.

4 Theoretical Analysis: RSP as Attention-Mediated Exploration

We isolate a transformer-level mechanism explaining the empirical regularities of Section 3: RSP affects the hidden state before the output softmax, its influence is controlled by the attention mass assigned to the random tokens, and this influence is structurally diluted as real-token evidence accumulates. The argument is built from standard self-attention identities; in particular, none of the formal results in Sections 4.3–4.5 assume $\mathbf{K} = \mathbf{V}$, weight-tied layers, or the existence of a global energy.

4.1 What RSP Is: Scale-Matched Isotropic Perturbation

Definition 1 (Random Soft Prompt)**.** *Let* $\mathbf{W}_E \in \mathbb{R}^{V \times d}$ *be the pretrained input-embedding table, with* μ_E *and* σ_E *the entrywise scalar mean and standard deviation defined in Section 3.1. An* RSP *of length* k *is a sequence* $(h_1, \ldots, h_k)$ *with* $h_j = \mu_E \mathbf{1}_d + \sigma_E\, \xi_j$ *and* $\xi_j \overset{iid}{\sim} \mathcal{N}(0, \mathbf{I}_d)$*. The centered prompt* $\bar{h}_j := h_j - \mu_E \mathbf{1}_d$ *satisfies* $\bar{h}_j \sim \mathcal{N}(0, \sigma_E^2 \mathbf{I}_d)$*. Whenever we analyze repeated-sampling protocols such as Pass@N, we additionally assume the prompt draws and any decoding randomness used across compared trials are independent.*

This is not a model of the empirical embedding distribution, which is generally anisotropic [Mu and Viswanath, 2018, Ethayarajh, 2019]. RSP deliberately discards directional geometry and replaces the row-covariance of $\mathbf{W}_E$ by the isotropic $\sigma_E^2 \mathbf{I}_d$: σ_E^2 borrows only the *scale*, so that the perturbation operates in the magnitude regime the model expects, while the identity covariance *deliberately* discards direction. The reason is that the branch-sensitive direction at a decoding state depends on prompt, layer, position, and the specific token pair compared; a generic exploration perturbation should not privilege a small set of embedding directions.

4.2 Why Randomness Is Necessary for Exploration

RSP is not random because we believe the embedding table is Gaussian; it is random because exploration requires resampling, centering, and direction-agnostic coverage. A fixed continuous prompt can steer a model, but reusing the same vector across rollouts produces the same perturbation direction every time, so it cannot generate independent alternatives for repeated sampling. Discrete random tokens are also inadequate, because each vocabulary token carries a lexical and semantic direction inherited from pretraining: discrete sampling injects uncontrolled semantic priors. RSP avoids both by using continuous vectors that are centered and isotropic rather than tied to any specific embedding direction. We formalize the choice through two properties.

Lemma 1 (Semantic neutrality)**.** *Let $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ and $\hat{h} := \bar{h}/\|\bar{h}\|$. For any fixed unit vectors $u_1, \ldots, u_M \in \mathbb{S}^{d-1}$ and $\delta \in (0, 1)$,*

$$\Pr\left[\max_{r \in [M]} |\hat{h}^\top u_r| \leq \sqrt{\tfrac{2\log(2M/\delta)}{d-1}}\right] \;\geq\; 1 - \delta.$$

A token embedding does not have this property: any fixed direction $u = e_s/\|e_s\|$ gives $|\hat{e}_s^\top u| = 1$. Lemma 1 therefore says, precisely, that an isotropic random direction is nearly orthogonal to any fixed finite collection of semantic anchors with high probability; it does *not* claim that a single random vector is free of every incidental correlation.

Theorem 1 (Direction-agnostic coverage)**.** *Let D be any zero-mean perturbation distribution on $\mathbb{R}^d$ with covariance Σ_D and trace budget $\operatorname{tr}(\Sigma_D) \leq \rho^2$. Define the worst directional variance $\mathcal{E}(D) := \min_{\|b\|=1} \operatorname{Var}_{h \sim D}(b^\top h)$. Then*

$$\mathcal{E}(D) \;=\; \lambda_{\min}(\Sigma_D) \;\leq\; \frac{\rho^2}{d},$$

with equality if and only if $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.

Theorem 1 is a minimax statement: among all zero-mean perturbations with the same total variance budget, isotropic covariance uniquely maximizes the least-covered direction. Together with Lemma 1 this rules out the most natural alternatives: a fixed prompt has $\Sigma_D = 0$ and produces no exploration; PCA-aligned noise is anisotropic and underexplores the orthogonal complement; vocabulary-token sampling is both anisotropic and semantically committing. The same theorem also rules out the apparently natural alternative of *matching* the empirical embedding covariance.

Corollary 1 (Embedding-matched noise is suboptimal)**.** *Let Σ_E be the row-covariance of $\mathbf{W}_E$ around the same scalar-broadcast center $\mu_E \mathbf{1}_d$ used in Definition 1, with $\operatorname{tr}(\Sigma_E) = d\sigma_E^2$. Compare two zero-mean perturbations on $\mathbb{R}^d$ with the common trace budget $\rho^2 := d\sigma_E^2$: the isotropic RSP law D_{iso} with covariance $\sigma_E^2 \mathbf{I}_d$, and the embedding-matched law D_{emb} with covariance Σ_E. Then $\mathcal{E}(D_{\text{iso}}) = \sigma_E^2 \geq \lambda_{\min}(\Sigma_E) = \mathcal{E}(D_{\text{emb}})$, with strict inequality whenever Σ_E is non-isotropic.*

The corollary requires no separate proof beyond Theorem 1. To the extent that pretrained embedding tables inherit the documented anisotropy of word and contextual representations [Mu and Viswanath, 2018, Ethayarajh, 2019], $\lambda_{\min}(\Sigma_E)$ is strictly smaller than σ_E^2, so an embedding-matched alternative leaves at least one direction less explored than the isotropic comparator. Discarding the directional geometry of $\mathbf{W}_E$ while keeping its scale is therefore not a simplification but a deliberately better choice for direction-agnostic exploration.

4.3 Exact Transformer-Level Mechanism

We now show that the effect of RSP in a standard transformer is controlled by the attention mass assigned to the random tokens. None of the statements in this subsection assume $\mathbf{K} = \mathbf{V}$, weight-tied layers, or the existence of a global energy function; they follow from standard self-attention identities. (A complementary interpretation as iterative retrieval on a Hopfield-style energy landscape [Ramsauer et al., 2021] is given in Appendix D; the formal results below do not depend on that view.)

Attention decomposition. Consider a self-attention layer ℓ with n real tokens and k RSP tokens. Let $\alpha_{ij}^{(\ell)}$ denote the attention weight from query position i to token j. Define the random-token attention mass

$$w_{r,i}^{(\ell)} \;=\; \sum_{j=1}^{k} \alpha_{i,n+j}^{(\ell)}, \tag{2}$$

and write $\mathbf{v}_j^{(\ell)}$ and $\mathbf{v}_{r_j}^{(\ell)}$ for the value vectors of the j-th real and the j-th RSP token, respectively. The attention output decomposes *exactly* as

$$\mathbf{x}_i^{(\ell+1)} \;=\; (1 - w_{r,i}^{(\ell)})\, \tilde{\mathbf{x}}_i^{(\ell+1)} \;+\; \boldsymbol{\eta}_i^{(\ell)}, \tag{3}$$

where $\tilde{\mathbf{x}}_i^{(\ell+1)} = \sum_{j=1}^{n} \frac{\alpha_{ij}^{(\ell)}}{1-w_{r,i}^{(\ell)}} \mathbf{v}_j^{(\ell)}$ is the renormalized attention output over the real tokens alone and $\boldsymbol{\eta}_i^{(\ell)} = \sum_{j=1}^{k} \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}$ is the RSP contribution. By the triangle inequality, $\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j\in[k]} \|\mathbf{v}_{r_j}^{(\ell)}\|$. Hence the same scalar $w_{r,i}^{(\ell)}$ controls both how much real-token attention is displaced and how large the random contribution can be. Eq. (3) is an algebraic identity, not an approximation; the proof and the deterministic bound on $\boldsymbol{\eta}_i^{(\ell)}$ are in Appendix E.1.

Sequence-wise dilution. With $\Delta_i^{(\ell)} := \min_{j\in[n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j'\in[k]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$ the real-vs-random pre-scaling logit gap, the random-token mass satisfies

$$w_{r,i}^{(\ell)} \;\leq\; \frac{k}{k + n\, \exp(\Delta_i^{(\ell)}/\sqrt{d})}. \tag{4}$$

For fixed k and $\Delta_i^{(\ell)}$ the right-hand side is strictly decreasing in n, so autoregressive cache growth alone imposes a monotone decrease on the largest random-token attention compatible with that gap. This is a structural dilution effect, not monotone decay of the realized $w_{r,i}^{(\ell)}$ on every path: $\Delta_i^{(\ell)}$ itself evolves with state, and layer-wise decay (deeper layers widen $\Delta_i^{(\ell)}$ as real-token retrieval sharpens) is a model-side hypothesis verified empirically by Figure 3.

Block-level propagation. A full transformer block adds residual connections, normalization, and a feed-forward sublayer. Under a common local Lipschitz constant for the full block map (covering the attention sublayer, LayerNorm, and the FFN) on a ball containing both the perturbed and unperturbed trajectories—which holds in practice because each component is locally Lipschitz on bounded sets [Xiong et al., 2020, Kim et al., 2021]—Appendix E.3 shows that the per-block perturbation amplitude scales linearly with $w_{r,i}^{(\ell)}$, so iterating across a finite depth gives a finite cumulative bound. Combined with Eq. (4), this explains why the perturbation enters strongly at early blocks and shrinks as evidence accumulates, without requiring a persistent perturbation throughout the reasoning chain.

Empirical verification. Figure 3 verifies the predicted layer-wise and sequence-wise decay of $w_{r,i}^{(\ell)}$ on Qwen2.5-Math-7B with suffix injection: attention to RSP tokens decays across both axes, while attention to question tokens remains comparatively uniform.

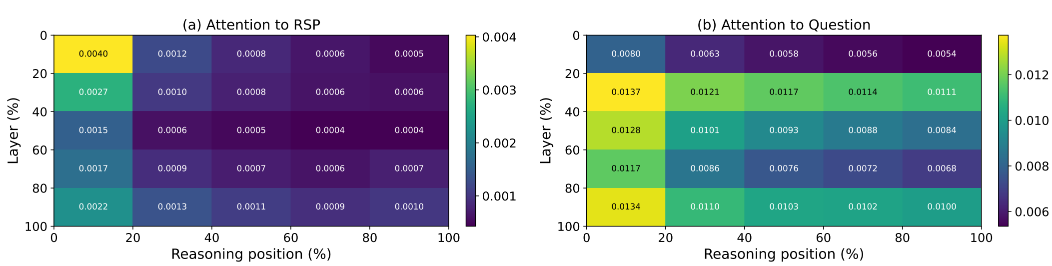

Figure 3: Per-token attention mass on Qwen2.5-Math-7B (suffix, 500 MATH-500 problems, each with an independently sampled RSP). Layer and reasoning-position axes are partitioned into five quantile bins. Attention to RSP tokens in (a) decays along both axes; attention to question tokens in (b) remains comparatively uniform.

4.4 Output Support Expansion Beyond Temperature

Temperature sampling is monotone, $p_i^{(\tau)} \propto p_i^{1/\tau}$, so $p_i > p_j \iff p_i^{(\tau)} > p_j^{(\tau)}$ for any $\tau > 0$: it cannot change token rankings at a fixed decoding state. RSP acts before the output softmax. Throughout this subsection $\bar{h} \in \mathbb{R}^d$ denotes one centered prompt vector from Definition 1, treated as a per-token surrogate; the multi-token compounding through softmax-weighted attention is handled empirically by Section 3.4. The two propositions below are stated for the local affine surrogate $z_i^{\mathrm{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}$ with $a_i := \nabla_{\bar{h}} z_i(0)$, valid for small $\|\bar{h}\|$ (Appendix E); they identify the structure of pre-softmax rank flips under that surrogate, while the multi-token, fully nonlinear effect is reported empirically in Section 3.4.

Proposition 1 (Stochastic ranking under the local affine surrogate)**.** *If there exists a token pair (i, j) with $a_i \neq a_j$ and the centered prompt law assigns positive probability to every nonempty open subset of $\mathbb{R}^d$, then $0 < \Pr_{\bar{h}}(z_i^{\mathrm{lin}}(\bar{h}) > z_j^{\mathrm{lin}}(\bar{h})) < 1$.*

Proposition 2 (Top-K entry under the local affine surrogate)**.** *Let $J_K(s_t)$ be the baseline top-K token set at decoding state s_t and $i \notin J_K(s_t)$. Define the surrogate entry region*

$$\mathcal{G}_{i,K} := \big\{\bar{h} \in \mathbb{R}^d :\ \#\{j \neq i : z_j^{\mathrm{lin}}(\bar{h}) > z_i^{\mathrm{lin}}(\bar{h})\} \leq K - 1\big\}.$$

A sufficient condition for $\mathcal{G}_{i,K}$ to contain a nonempty open set is the existence of $\bar{h}_0 \in \mathbb{R}^d$ at which token i outranks all but at most $K-1$ other tokens with a strict margin *(no tie); the affine surrogate is continuous, so an open ball around such $\bar{h}_0$ lies in $\mathcal{G}_{i,K}$ (Appendix E). If this condition holds and the centered prompt law assigns positive probability to every nonempty open subset of $\mathbb{R}^d$, then* $\Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0$.

Proposition 2 is intentionally conditional: it does not claim every excluded token admits an entry region. The complementary case in which token i is dominated by the baseline top-K in the affine surrogate is exactly the case in which the proposition makes no claim. Temperature scaling reaches no $\mathcal{G}_{i,K}$ at all because it never changes token order; the surrogate-level mechanism here is branch opening, not probability flattening.

4.5 From Independent Perturbations to Pass@N

Let $\mu_i := \Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0$.

Corollary 2 (Sample complexity of admission)**.** *Within N independent fresh RSP draws,* $\Pr($*at least one admits i into top-K*$) = 1 - (1-\mu_i)^N \geq 1 - \exp(-N\mu_i)$.

Lifting this to task accuracy requires an additional task-dependent assumption.

Proposition 3 (Exploration-meets-multi-path accuracy gain)**.** *Fix a problem x. Run the deterministic baseline once and draw N fresh independent RSPs, decoding once per sample. Let $\mathcal{T}^\star(x)$ be the nonempty set of correct trajectories and $p_{\mathrm{base}}(x) \in \{0, 1\}$ the baseline correctness indicator. Assume* (M1) *for each n, the marginal probability that the n-th RSP run produces a trajectory in $\mathcal{T}^\star(x)$ is at least $p_{\min}(x) > 0$, and* (M2) *the N RSP runs are mutually independent. Then* $\Pr\big(\mathrm{Pass@}(N{+}1)$ *correct on* $x\big) \geq \max\big\{p_{\mathrm{base}}(x),\ 1 - (1 - p_{\min}(x))^N\big\} \geq 1 - e^{-Np_{\min}(x)}$.

(M1) is task-side; Section 3.4 reports the empirical signature, with RSP-induced trajectories including both improvements and regressions and net gain under repeated sampling.

5 Conclusion

We studied Random Soft Prompts (RSPs), isotropic Gaussian vectors fitted to the entrywise mean and variance of the pretrained embedding table and injected without training. Empirically RSPs raise early-stage entropy while later generation stabilizes (Section 3); theoretically the RSP contribution at each attention layer is exactly captured by a scalar attention mass that cache growth structurally dilutes, and under a local affine analysis RSPs can admit previously excluded tokens into the effective top-K candidate set in a way temperature scaling cannot (Section 4). Because RSP is rank-changing while temperature is rank-preserving, the two are complementary, suggesting a sampling-level remedy for entropy collapse in GRPO [Shao et al., 2024] beyond DAPO [Yu et al., 2025] and CE-GPPO [Su et al., 2026]; training-time validation, extension beyond mathematical reasoning, and a principled choice of RSP length and injection position—together with the task-side bound $p_{\min}(x)>0$ of Proposition 3—are natural next steps.

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A Experimental Setup and Full Accuracy Breakdown

Table 5 provides the full per-position accuracy breakdown (Prefix, Infix, Suffix) corresponding to the best-across-positions results summarized in Table 1 in the main text. Table 6 lists the number of RSP tokens L used for each model–benchmark pair, selected from $\{10, 15, 20\}$. All experiments are conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-Math-1.5B variants).

Table 5: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Mode	MATH-500	GSM8K	AIME24
Instruct Models				
LLaMA-3.1-8B-Instruct	Baseline	**52.20**	85.75	6.67
	Prefix	45.00 (−7.2)	76.35 (−9.4)	3.33 (−3.3)
	Infix	49.40 (−2.8)	85.97 (+0.2)	**10.00** (+3.3)
	Suffix	49.40 (−2.8)	**86.73** (+1.0)	**10.00** (+3.3)
Qwen2.5-Math-7B-Instruct	Baseline	83.20	95.45	13.33
	Prefix	81.40 (−1.8)	94.92 (−0.5)	13.33 (+0.0)
	Infix	**84.20** (+1.0)	95.60 (+0.2)	**16.67** (+3.3)
	Suffix	83.40 (+0.2)	**95.68** (+0.2)	**16.67** (+3.3)
Qwen2.5-Math-1.5B-Instruct	Baseline	73.00	85.29	10.00
	Prefix	74.80 (+1.8)	85.29 (+0.0)	13.33 (+3.3)
	Infix	**75.20** (+2.2)	**85.75** (+0.5)	6.67 (−3.3)
	Suffix	74.20 (+1.2)	84.69 (−0.6)	**20.00** (+10.0)
Base Models (Raw Text)				
Qwen2.5-Math-7B	Baseline	**72.20**	84.15	6.67
	Prefix	71.60 (−0.6)	**84.31** (+0.2)	**16.67** (+10.0)
	Infix	63.20 (−9.0)	64.29 (−19.9)	**16.67** (+10.0)
	Suffix	55.60 (−16.6)	54.13 (−30.0)	13.33 (+6.7)
Qwen2.5-Math-1.5B	Baseline	61.40	**77.86**	**16.67**
	Prefix	**64.40** (+3.0)	74.30 (−3.6)	10.00 (−6.7)
	Infix	63.20 (+1.8)	73.92 (−3.9)	10.00 (−6.7)
	Suffix	54.60 (−6.8)	58.61 (−19.3)	3.33 (−13.3)
Base Models + ChatML (Format Mismatch)				
Qwen2.5-Math-7B	Baseline	52.20	58.30	**23.33**
	Prefix	59.20 (+7.0)	56.41 (−1.9)	3.33 (−20.0)
	Infix	62.00 (+9.8)	69.07 (+10.8)	**23.33** (+0.0)
	Suffix	**70.40** (+18.2)	**75.97** (+17.7)	**23.33** (+0.0)
Qwen2.5-Math-1.5B	Baseline	34.40	37.45	13.33
	Prefix	**63.40** (+29.0)	**67.02** (+29.6)	**20.00** (+6.7)
	Infix	39.20 (+4.8)	50.42 (+13.0)	6.67 (−6.7)
	Suffix	51.00 (+16.6)	50.64 (+13.2)	13.33 (+0.0)

Table 6: Number of RSP tokens (L) used for each model and benchmark.

Model	MATH-500	GSM8K	AIME24
LLaMA-3.1-8B-Instruct	15	20	15
Qwen2.5-Math-7B-Instruct	20	20	20
Qwen2.5-Math-1.5B-Instruct	20	10	20
Qwen2.5-Math-7B	10	10	20
Qwen2.5-Math-1.5B	10	10	20
Qwen2.5-Math-1.5B (ChatML)	20	20	10
Qwen2.5-Math-7B (ChatML)	20	20	20

A.1 RSP Injection Positions and Prompt Templates

Below we show the prompt structure for each injection position across all model types. Blue text indicates RSP embeddings, and purple text indicates special tokens.

A.1.1 Prefix

RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

ChatML (Qwen Instruct / Base + ChatML).

```
[Random Embeddings (L tokens)]
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
[Random Embeddings (L tokens)]
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
[Random Embeddings (L tokens)]
{question}
Please reason step by step, and put your final answer within \boxed{}.
```

A.1.2 Infix

A description text and special tokens are inserted into the prompt. The special token positions are then replaced with RSP embeddings at the embedding level.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens: <special_token_1>...<special_token_L>
<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
```

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens:
<reserved_special_token_0>...
<reserved_special_token_L>
<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}
There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.
Here are the {L} special tokens: <special_token_1>...<special_token_L>
Please reason step by step, and put your final answer within \boxed{}.
```

A.1.3 Suffix

For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token. For raw text models, RSP embeddings are concatenated at the end of the prompt.

ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{}.<|im_end|>
<|im_start|>user
{question}[Random Embeddings (L tokens)]<|im_end|>
<|im_start|>assistant
```

LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{}.<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}[Random Embeddings (L tokens)]<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

Raw Text (Qwen Base).

```
{question}
Please reason step by step, and put your final answer within \boxed{}.[Random
Embeddings (L tokens)]
```

A.2 Answer Extraction and Grading

The final answer is extracted from the model output through a fallback chain. Each step is attempted in order; if extraction fails, the next step is tried.

Answer Extraction Fallback Chain

1. `"final answer is $...$. I hope"` pattern (Minerva format)
2. `\boxed{...}`: last non-empty box is used; empty `\boxed{}` are skipped
3. Text following `"the answer is"`
4. Text following `"final answer is"`
5. Last number in the output (regex fallback)

Extracted answers are normalized by removing `$`, `\left`, `\right`, and unit strings. Grading is performed via symbolic equivalence checking: exact string match, numerical comparison (`math.isclose`, `rel_tol=1e-4`), and SymPy symbolic simplification.

A.3 Reproduced Prior Work Hyperparameters

All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct, evaluated with greedy decoding (temperature $= 0$) and our unified answer extraction pipeline.

TTSV. Prefix length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

LTPO. Table 7 reports the per-benchmark hyperparameters.

Table 7: LTPO hyperparameters.

Parameter	GSM8K	MATH-500	AIME24
tokens	8	8	8
steps	20	20	20
top_k	10	10	10
sigma	20	20	5
sigma_decay	0.95	0.95	0.95
lr	1e-2	5e-3	5e-2

SoftCoT. Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

A.4 Full Entropy Curves

Figure 4 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy, and Table 8 reports the corresponding scalar statistics including overall means, first-5% means, and the average number of generated tokens per problem. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

Table 8: Per-step statistics for Qwen2.5-Math-1.5B-Instruct on MATH-500 across the full generation and the first 5% of generation steps, together with the average number of generated tokens per problem. Top-1 Prob (overall) is reported as a weighted average over the first-10%/middle/last-10% segment means with weights 0.1/0.8/0.1. Extension of Figure 1 (main text) and Figure 4.

Metric	Baseline	Prefix	Infix	Suffix	LTPO	SoftCoT	TTSV
Tokens (mean)	575.7	558.4	549.9	573.3	592.3	594.4	577.4
Entropy (first 5%)	0.1332	0.1366	0.1402	0.1941	0.1662	0.4218	0.1359
Entropy (overall)	0.0871	0.0881	0.0899	0.1049	0.0909	0.1059	0.0864
Top-1 Prob (first 5%)	0.9535	0.9523	0.9525	0.9373	0.9437	0.9156	0.9534
Top-1 Prob (overall)	0.9684	0.9681	0.9678	0.9647	0.9683	0.9651	0.9685
Varentropy (first 5%)	0.2181	0.2207	0.2463	0.4010	0.2953	2.2234	0.2174
Varentropy (overall)	0.1353	0.1375	0.1422	0.2038	0.1452	0.2404	0.1361

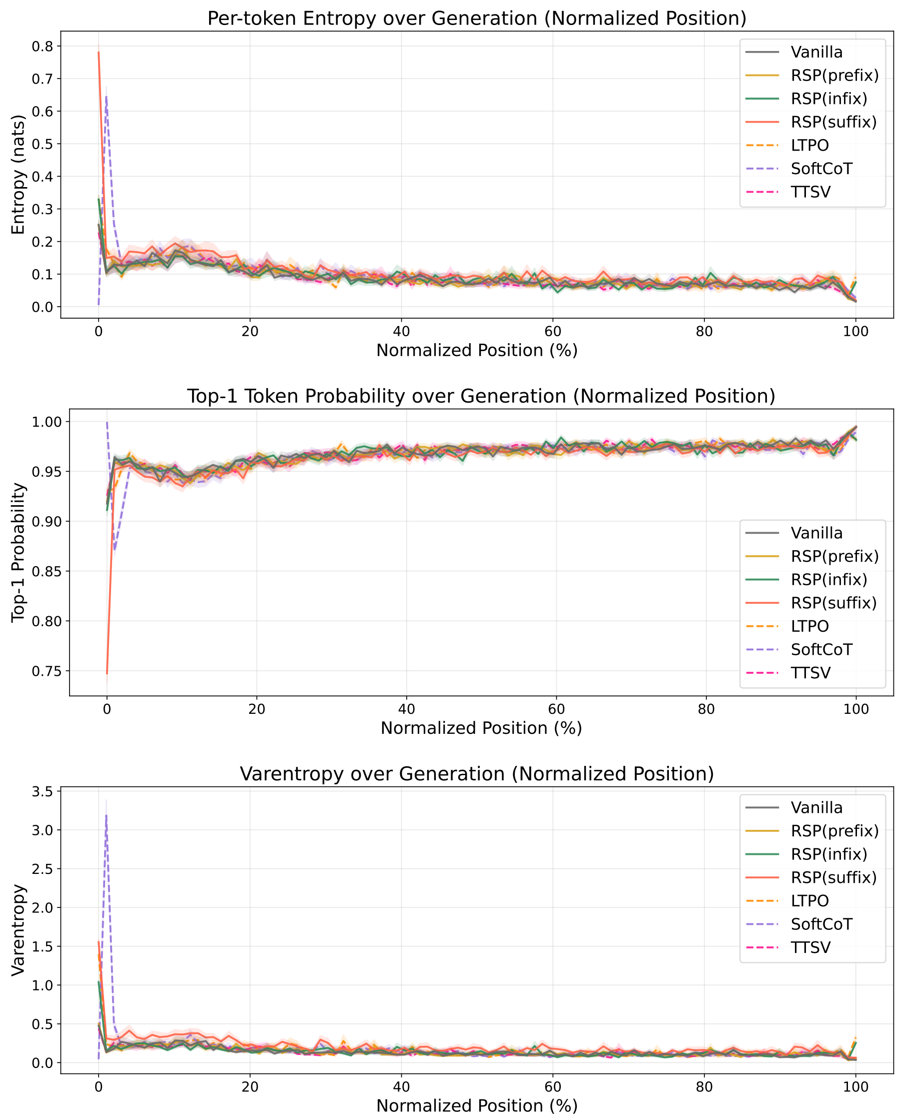


Figure 4: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods. All methods converge after the initial stage.

B Token-Level Distribution Metrics

This appendix provides formal definitions for the three metrics used in the Token-level analysis of Section 3.4. Let P_{base} denote the baseline next-token distribution at $\tau = 1.0$ and P_{target} the candidate distribution being compared (e.g., the baseline at $\tau = 2.0$ or RSP at $\tau = 1.0$). All metrics are computed analytically from saved logits at the first generation step.

Spearman rank correlation. Spearman's ρ is the Pearson correlation coefficient between the token ranks of two distributions:

$$\rho \;=\; \mathrm{Pearson}(\mathrm{rank}(P_{\text{target}}),\; \mathrm{rank}(P_{\text{base}}))\,. \tag{5}$$

Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore indicates rank reorganization beyond what temperature scaling can produce.

Mass outside top-K. Let $\mathrm{TopK}(P_{\text{base}})$ be the set of K tokens with the highest probability under P_{base}. The probability mass that the candidate distribution places *outside* this set is

$$\mathrm{MassOutside}_K \;=\; 1 - \sum_{t \in \mathrm{TopK}(P_{\text{base}})} P_{\text{target}}(t). \tag{6}$$

This directly measures support expansion: how much probability the candidate assigns to tokens that are not in the baseline's preferred set. Reported values use $K = 10$.

Jensen–Shannon divergence. The symmetrized, bounded version of Kullback–Leibler divergence:

$$\mathrm{JS}(P, Q) \;=\; \tfrac{1}{2}\,\mathrm{KL}(P \,\|\, M) + \tfrac{1}{2}\,\mathrm{KL}(Q \,\|\, M), \qquad M = \tfrac{1}{2}(P + Q). \tag{7}$$

We use base-2 logarithms so that JS $\in [0, 1]$. Tokens absent from the stored top-100 are assigned a sentinel logit (-10^9) before softmax, which is negligible for our setting since the top-5 covers more than 99.9% of the probability mass.

C Why Random Prompts Are Semantically Neutral and Direction-Agnostic

This appendix formalizes the "why random" claim used in Section 4.2. The point is not that random prompts contain hidden task information. The point is structural: if the perturbation is meant to act as a generic exploration device rather than as a learned prompt prior, then it should be resampled across runs, centered so that it does not induce a fixed first-order drift, and direction-agnostic so that it does not under-serve whichever local branch direction matters for the current prompt.

C.1 Semantic Neutrality

Let $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ and $\hat{h} := \bar{h}/\|\bar{h}\|$ whenever $\bar{h} \neq 0$. Let $u_1, \ldots, u_M \in \mathbb{S}^{d-1}$ be any fixed unit vectors representing semantic anchor directions.

Lemma 2 (Semantic neutrality). *For any $\delta \in (0,1)$,*

$$\Pr\left[\max_{r \in [M]} |\hat{h}^\top u_r| \leq \sqrt{\frac{2\log(2M/\delta)}{d-1}}\right] \geq 1 - \delta.$$

Hence the normalized direction of an isotropic Gaussian prompt is simultaneously nearly orthogonal to every fixed finite collection of semantic anchors with high probability.

Proof. Because $\bar{h}$ is isotropic Gaussian, its normalized direction $\hat{h}$ is uniformly distributed on the sphere $\mathbb{S}^{d-1}$. For a fixed unit vector $u \in \mathbb{S}^{d-1}$, the marginal $\hat{h}^\top u$ satisfies the standard spherical concentration bound [Vershynin, 2018, Theorem 3.4.6][Boucheron et al., 2013, Section 3.2],

$$\Pr\left(|\hat{h}^\top u| \geq t\right) \leq 2\exp\left(-\frac{(d-1)t^2}{2}\right) \qquad \text{for } t \in (0,1),$$

which follows from the spherical-cap area bound $\sigma\{\hat{h} \in \mathbb{S}^{d-1} : \hat{h}^\top u \geq t\} \leq \frac{1}{2}(1-t^2)^{(d-1)/2}$ together with $(1-t^2)^{(d-1)/2} \leq \exp(-(d-1)t^2/2)$ derived from $\log(1-x) \leq -x$ for $x \in [0,1)$. Applying this bound to each u_r and using a union bound,

$$\Pr\left[\max_{r \in [M]} |\hat{h}^\top u_r| \geq t\right] \leq 2M\exp\left(-\frac{(d-1)t^2}{2}\right).$$

Setting

$$t = \sqrt{\frac{2\log(2M/\delta)}{d-1}}$$

gives the claim. If the displayed value of t exceeds 1, the statement is trivial because $|\hat{h}^\top u_r| \leq 1$ always. $\square$

The contrast with a token embedding is immediate. If $e_s \neq 0$ is a fixed token embedding and $\hat{e}_s := e_s/\|e_s\|$, then choosing the anchor direction $u = \hat{e}_s$ gives

$$|\hat{e}_s^\top u| = 1.$$

A token prompt is therefore maximally aligned with at least one semantic direction, whereas an isotropic random prompt is nearly orthogonal to every fixed finite set of such directions. This is the sense in which tokens inject semantic commitment while random prompts act as semantically uncommitted perturbations.

C.2 Centered Perturbations Do Not Impose Systematic First-Order Drift

Let Δ_{ij} be the baseline logit gap between tokens i and j, and write the first-order surrogate as

$$\Delta_{ij}(\bar{h}) = \Delta_{ij} + b_{ij}^\top \bar{h}.$$

Proposition 4 (No systematic first-order drift). *If $\mathbb{E}[\bar{h}] = 0$, then*

$$\mathbb{E}\big[\Delta_{ij}(\bar{h})\big] = \Delta_{ij}$$

for every token pair (i,j). If a deterministic prompt h_0 is reused across runs, then

$$\Delta_{ij}(h_0) - \Delta_{ij} = b_{ij}^\top h_0$$

is a fixed directional bias.

Proof. The centered case is immediate from linearity of expectation:

$$\mathbb{E}\big[\Delta_{ij}(\bar{h})\big] = \Delta_{ij} + b_{ij}^\top \mathbb{E}[\bar{h}] = \Delta_{ij}.$$

The deterministic case follows by substitution. □

The proposition does not say that a centered perturbation leaves every *realization* unchanged; it says there is no systematic first-order push toward one branch across runs. This is exactly the behavior desired from a generic exploration device. A fixed token or fixed soft prompt may still be useful, but it acts as a persistent prompt prior rather than as an unbiased exploration mechanism.

C.3 Direction-Agnostic Coverage

The second requirement is directional coverage. Let D be any zero-mean perturbation distribution in $\mathbb{R}^d$ with covariance Σ_D and variance budget $\operatorname{tr}(\Sigma_D) \le \rho^2$. For a unit vector b, the first-order perturbation energy available along b is $\operatorname{Var}_{h\sim D}(b^\top h) = b^\top \Sigma_D b$. This is the quantity that appears in the local linear logit surrogate used in the main text.

Theorem 2 (Direction-agnostic coverage)**.** *Define*

$$\mathcal{E}(D) := \min_{\|b\|=1} \operatorname{Var}_{h\sim D}(b^\top h).$$

Then

$$\mathcal{E}(D) = \lambda_{\min}(\Sigma_D) \le \frac{\rho^2}{d},$$

with equality if and only if $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.

Proof. For any symmetric covariance matrix Σ_D,

$$\min_{\|b\|=1} b^\top \Sigma_D b = \lambda_{\min}(\Sigma_D),$$

so $\mathcal{E}(D) = \lambda_{\min}(\Sigma_D)$. Let the eigenvalues of Σ_D be $\lambda_1, \ldots, \lambda_d$. Then

$$\lambda_{\min}(\Sigma_D) \le \frac{1}{d}\sum_{m=1}^{d} \lambda_m = \frac{\operatorname{tr}(\Sigma_D)}{d} \le \frac{\rho^2}{d}.$$

Equality holds if and only if all eigenvalues are equal, i.e., $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. □

This theorem is a statement about second moments, not about Gaussianity by itself. Many distributions could realize isotropic covariance. We use a Gaussian in RSP because it is the simplest centered, rotation-symmetric family that matches the scale of pretrained embeddings. The theorem clarifies the structural advantage: a fixed prompt has $\Sigma_D = 0$ and therefore no across-run exploration, whereas any anisotropic perturbation leaves at least one branch-sensitive direction less explored than the isotropic optimum.

Proposition 5 (Covariance pushforward under the value projection)**.** *Let $\bar{h}$ be a centered random vector with covariance $\sigma_E^2\mathbf{I}_d$, and let $v = W\bar{h}$ for a matrix $W \in \mathbb{R}^{d_v\times d}$. Then*

$$\operatorname{Cov}(v) = \sigma_E^2 W W^\top.$$

Consequently, for every unit vector $u \in \mathbb{S}^{d_v-1}$,

$$\sigma_E^2 \lambda_{\min}(WW^\top) \le \operatorname{Var}(u^\top v) = \sigma_E^2 u^\top W W^\top u \le \sigma_E^2 \lambda_{\max}(WW^\top).$$

In particular, if W is full rank then the projected perturbation has strictly positive variance in every output-space direction.

Proof. Since $v = W\bar{h}$ and $\mathbb{E}[\bar{h}] = 0$,

$$\operatorname{Cov}(v) = \mathbb{E}[vv^\top] = W\,\mathbb{E}[\bar{h}\bar{h}^\top]\,W^\top = \sigma_E^2 W W^\top.$$

For any unit vector u,

$$\operatorname{Var}(u^\top v) = u^\top \operatorname{Cov}(v)\, u = \sigma_E^2 u^\top W W^\top u,$$

and the Rayleigh quotient bounds give the stated inequalities. □

D Interpretive Connection to Hopfield Retrieval (Optional)

This appendix is included for completeness and is not load-bearing for any quantitative result in the main text. Section 4.3 of the main paper deliberately uses only standard self-attention identities (the exact decomposition of Eq. (3) and the decay bound of Eq. (4)); the formal claims there do not assume $\mathbf{K} = \mathbf{V}$, weight-tied layers, or the existence of a global energy. The present appendix records the alternative reading—originating from the Modern Hopfield correspondence [Ramsauer et al., 2021]—under which a single attention update with $\mathbf{K} = \mathbf{V}$ becomes one gradient step on a Hopfield energy. This reading provides additional intuition for why a deterministic forward pass can be exploration-poor and has been a useful lens in the related literature, but it is not used in any proof of Section 4.

We restate the Modern Hopfield correspondence in the form used in this interpretive lens. For a key matrix $\mathbf{K} = (\mathbf{k}_1, \ldots, \mathbf{k}_n)$ at a fixed query position, the Hopfield energy is

$$E_{\mathbf{K}}(\sigma) := -\beta^{-1} \log \sum_{j=1}^{n} \exp(\beta\, \mathbf{k}_j^\top \sigma) + \tfrac{1}{2}\|\sigma\|^2.$$

A direct calculation gives the gradient

$$\nabla E_{\mathbf{K}}(\sigma) = \sigma - \mathbf{K}^\top \mathrm{softmax}(\beta \mathbf{K}\sigma).$$

Under (A1) $\mathbf{K} = \mathbf{V}$, the attention update with this $\mathbf{K}$ is therefore $\sigma^{\mathrm{new}} = \sigma - \nabla E_{\mathbf{K}}(\sigma) = \mathbf{K}^\top \mathrm{softmax}(\beta \mathbf{K}\sigma)$, which is one gradient step on $E_{\mathbf{K}}$ at unit step size [Ramsauer et al., 2021].

Lemma 3 (Descent inequality). *If $E_{\mathbf{K}}$ is L_E-smooth on a neighborhood of σ, then the unit-step gradient update $\sigma^{new} := \sigma - \nabla E_{\mathbf{K}}(\sigma)$ satisfies*

$$E_{\mathbf{K}}(\sigma^{new}) \le E_{\mathbf{K}}(\sigma) - \big(1 - \tfrac{L_E}{2}\big)\|\nabla E_{\mathbf{K}}(\sigma)\|^2.$$

Proof. The standard L_E-smooth quadratic upper bound [Nesterov, 2018, Lemma 1.2.3] states $E(\sigma + v) \le E(\sigma) + \nabla E(\sigma)^\top v + (L_E/2)\|v\|^2$ for any v in the smoothness neighborhood. Applying this to $v = -\nabla E(\sigma)$ gives $E(\sigma - \nabla E) \le E(\sigma) - \|\nabla E\|^2 + (L_E/2)\|\nabla E\|^2 = E(\sigma) - (1 - L_E/2)\|\nabla E\|^2$. □

Theorem 3 (Deterministic descent and sublevel-set invariance). *Assume* (A1) $\mathbf{K} = \mathbf{V}$, (A2) *the attention component is weight-tied across blocks, and* (A3) *$E_{\mathbf{K}}$ is L_E-smooth on a neighborhood of the unperturbed trajectory with $L_E < 2$. Then the unit-step iterate $\sigma^{(l+1)} = \sigma^{(l)} - \nabla E_{\mathbf{K}}(\sigma^{(l)})$ is strictly monotone-descending in $E_{\mathbf{K}}$ at every non-stationary point, and the sublevel set $\{\sigma : E_{\mathbf{K}}(\sigma) \le E_{\mathbf{K}}(\sigma^{(0)})\}$ is forward-invariant.*

Proof. By Lemma 3, $E_{\mathbf{K}}(\sigma^{(l+1)}) \le E_{\mathbf{K}}(\sigma^{(l)}) - (1 - L_E/2)\|\nabla E_{\mathbf{K}}(\sigma^{(l)})\|^2$. Under (A3), the coefficient $1 - L_E/2 > 0$, so $E_{\mathbf{K}}(\sigma^{(l+1)}) \le E_{\mathbf{K}}(\sigma^{(l)})$ with strict inequality unless $\nabla E_{\mathbf{K}}(\sigma^{(l)}) = 0$, which proves forward invariance. □

Corollary 3 (Connected-component containment for the unit-step iterate). *Let $S := \{\sigma : E_{\mathbf{K}}(\sigma) \le E_{\mathbf{K}}(\sigma^{(0)})\}$ and let $C(\sigma^{(0)})$ denote the connected component of S containing $\sigma^{(0)}$. Define*

$$\mathrm{sep}(C(\sigma^{(0)})) \;:=\; \inf\big\{\|x - y\| \;:\; x \in C(\sigma^{(0)}),\; y \in S \setminus C(\sigma^{(0)})\big\},$$

with the convention $\mathrm{sep} = +\infty$ *if S is connected. Assume* $\mathrm{sep}(C(\sigma^{(0)})) > 0$. *Then the unit-step iterate $\sigma^{(l+1)} = \sigma^{(l)} - \nabla E_{\mathbf{K}}(\sigma^{(l)})$ remains in $C(\sigma^{(0)})$ for all $l \ge 0$ whenever*

$$\|\nabla E_{\mathbf{K}}(\sigma^{(l)})\| \;<\; \mathrm{sep}(C(\sigma^{(0)})) \qquad \textit{for every } l.$$

Proof. By Theorem 3, every iterate lies in S. Suppose by induction $\sigma^{(l)} \in C(\sigma^{(0)})$. The displacement $\sigma^{(l+1)} - \sigma^{(l)} = -\nabla E_{\mathbf{K}}(\sigma^{(l)})$ has Euclidean length $\|\nabla E_{\mathbf{K}}(\sigma^{(l)})\|$. A jump to a distinct component $C' \subseteq S$ would require this length to be at least $\inf\{\|x-y\| : x \in C(\sigma^{(0)}),\, y \in C'\} \ge \mathrm{sep}(C(\sigma^{(0)}))$, which the hypothesis excludes. Hence $\sigma^{(l+1)} \in C(\sigma^{(0)})$. □

Positivity of sep. $\mathrm{sep}(C) > 0$ holds whenever the components of S are spatially separated by a region of strictly higher energy than $E_{\mathbf{K}}(\sigma^{(0)})$, which we treat as a problem-dependent assumption rather than a structural property: it requires that the relevant components of the Hopfield energy do not share closure boundary points at level $E_{\mathbf{K}}(\sigma^{(0)})$. If $\mathrm{sep} = 0$ the corollary becomes vacuous; in that case the trajectory is still in S by Theorem 3, but is not formally trapped in $C(\sigma^{(0)})$.

Caveats and scope. Theorem 3 depends on three assumptions that are individually mild but jointly idealized for real transformers. (A1) $\mathbf{K} = \mathbf{V}$ collapses the key and value projections; this is the standard Modern Hopfield assumption [Ramsauer et al., 2021] and isolates the energy-iteration aspect of attention from the additional value-projection degree of freedom. (A2) weight-tying treats the attention component as a single repeated map; without it, each block has its own energy and the trajectory does not minimize a single fixed function across blocks. (A3) is a smoothness condition on $E_{\mathbf{K}}$. For the Hopfield energy used here, $\nabla^2 E_{\mathbf{K}}(\sigma) = \mathbf{I} - \beta \mathbf{K}^\top (\mathrm{diag}(p_\sigma) - p_\sigma p_\sigma^\top)\mathbf{K}$ where $p_\sigma = \mathrm{softmax}(\beta \mathbf{K} \sigma)$, so the operator-norm bound on $\nabla^2 E_{\mathbf{K}}$ depends on the inverse-temperature β and on $\|\mathbf{K}\|$. (A3) holds whenever $\beta \|\mathbf{K}\|^2$ is sufficiently small, which is the standard regime in which the Hopfield iterate converges in a small number of steps. The theorem therefore characterizes an idealized tendency of attention-only iteration in this regime; it is not a literal property of every transformer block. In particular, sublevel-set confinement follows from Theorem 3, whereas connected-component confinement additionally requires the separation and per-step gradient assumptions of Corollary 3. Its role in the main text is to supply an idealized source of exploration deficit that an injected stochastic perturbation addresses; the empirical link is the cross-layer attention decay observed in Figure 3 and the trajectory diversity gain reported in Section 3.4.

E Proofs for the Transformer-Level Mechanism

This appendix collects the formal proofs underlying Sections 4.3–4.5 of the main text.

E.1 Exact Attention Decomposition and Magnitude Control

Proposition 6 (Attention decomposition)**.** *Let $\alpha_{ij}^{(l)}$ be the attention weights for query position i at layer l, with n real tokens and k random tokens. Define*

$$w_{r,i}^{(l)} := \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)}$$

and, for $j \in [n]$,

$$\tilde{\alpha}_{ij}^{(l)} := \frac{\alpha_{ij}^{(l)}}{1 - w_{r,i}^{(l)}}.$$

Then

$$\mathbf{x}_i^{(l+1)} = (1 - w_{r,i}^{(l)})\, \tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)},$$

where

$$\tilde{\mathbf{x}}_i^{(l+1)} := \sum_{j=1}^{n} \tilde{\alpha}_{ij}^{(l)} \mathbf{v}_j^{(l)} \qquad \textit{and} \qquad \boldsymbol{\eta}_i^{(l)} := \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}.$$

Proof. The attention output is

$$\mathbf{x}_i^{(l+1)} = \sum_{j=1}^{n} \alpha_{ij}^{(l)} \mathbf{v}_j^{(l)} + \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)}.$$

By definition, $\sum_{j=1}^{n} \alpha_{ij}^{(l)} = 1 - w_{r,i}^{(l)}$, so

$$\sum_{j=1}^{n} \alpha_{ij}^{(l)} \mathbf{v}_j^{(l)} = (1 - w_{r,i}^{(l)}) \sum_{j=1}^{n} \frac{\alpha_{ij}^{(l)}}{1 - w_{r,i}^{(l)}} \mathbf{v}_j^{(l)} = (1 - w_{r,i}^{(l)})\, \tilde{\mathbf{x}}_i^{(l+1)}.$$

Substituting this identity gives the decomposition. $\square$

Corollary 4 (Magnitude scales with random-token attention)**.** *For every realization of the random values,*

$$\|\boldsymbol{\eta}_i^{(l)}\| \le w_{r,i}^{(l)} \max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\|.$$

Proof. By the triangle inequality,

$$\|\boldsymbol{\eta}_i^{(l)}\| = \left\| \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \mathbf{v}_{r_j}^{(l)} \right\| \le \sum_{j=1}^{k} \alpha_{i,n+j}^{(l)} \|\mathbf{v}_{r_j}^{(l)}\| \le w_{r,i}^{(l)} \max_{j \in [k]} \|\mathbf{v}_{r_j}^{(l)}\|.$$

$\square$

Corollary 4 is the only quantitative fact needed in the main text. No independence assumption between attention weights and random keys is required. Once $w_{r,i}^{(l)}$ is small, the random contribution must be small as well.

E.2 Attention Decay Bound

Theorem 4 (Attention decay)**.** *The total attention on random tokens satisfies*

$$w_{r,i}^{(l)} \leq \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}, \tag{8}$$

where $\Delta_i^{(l)} := \min_{j\in[n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j'\in[k]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$*, is the real-vs-random logit gap.*

Proof. Let $s_j = \mathbf{q}_i^\top \mathbf{k}_j/\sqrt{d}$ for real keys and $s_{r_{j'}} = \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}/\sqrt{d}$ for random keys. By definition of $\Delta_i^{(l)}$, each real key satisfies $s_j \geq \Delta_i^{(l)}/\sqrt{d} + s_{r,\max}$ where $s_{r,\max} = \max_{j'} s_{r_{j'}}$. Therefore $\exp(s_j) \geq \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \exp(s_{r,\max})$ for every real key. Summing over n real keys gives

$$\sum_{j=1}^{n} \exp(s_j) \geq n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \exp(s_{r,\max}) \geq n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot \frac{1}{k}\sum_{j'=1}^{k} \exp(s_{r_{j'}}).$$

Let $R := \sum_{j'=1}^{k} \exp(s_{r_{j'}})$. Then

$$w_{r,i}^{(l)} = \frac{R}{\sum_{j=1}^{n} \exp(s_j) + R} \leq \frac{R}{n \cdot \exp(\Delta_i^{(l)}/\sqrt{d}) \cdot R/k + R} = \frac{k}{k + n \cdot \exp(\Delta_i^{(l)}/\sqrt{d})}.$$

□

Corollary 5 (Sequence-wise annealing under a fixed logit gap)**.** *For fixed k and $\Delta_i^{(l)}$, the upper bound*

$$f(n) = \frac{k}{k + n \exp(\Delta_i^{(l)}/\sqrt{d})}$$

is strictly decreasing in n. Consequently, cache growth alone decreases the maximum perturbation mass compatible with that gap.

Proof. Differentiate $f(n)$ with respect to n:

$$f'(n) = -\frac{k \exp(\Delta_i^{(l)}/\sqrt{d})}{\Big(k + n \exp(\Delta_i^{(l)}/\sqrt{d})\Big)^2} < 0.$$

Thus $f(n)$ is strictly decreasing. □

Corollary 6 (Vanishing random contribution under annealing)**.** *If $n \to \infty$ along decoding while k and $\Delta_i^{(l)}$ remain fixed, then $w_{r,i}^{(l)} \to 0$. Moreover, for any fixed realization of the k random values,*

$$\|\boldsymbol{\eta}_i^{(l)}\| \leq w_{r,i}^{(l)} \max_{j\in[k]} \|\mathbf{v}_{r_j}^{(l)}\| \to 0.$$

Since the implementation samples finite-dimensional vectors and k is finite, the displayed maximum is almost surely finite.

Proof. Corollary 5 implies $w_{r,i}^{(l)} \to 0$. The bound on $\|\boldsymbol{\eta}_i^{(l)}\|$ is Corollary 4. Since the multiplier $\max_{j\in[k]} \|\mathbf{v}_{r_j}^{(l)}\|$ is finite for any fixed realization of finitely many sampled values, the product converges to zero. □

E.3 Feed-Forward Propagation under Local Lipschitz Control

A transformer block applies an attention sublayer followed by a feed-forward sublayer with residual connections:

$$\mathbf{x}_i^{(l,\text{attn})} = \mathbf{x}_i^{(l)} + \mathrm{Attn}^{(l)}(\mathbf{x}_i^{(l)}), \qquad \mathbf{x}_i^{(l+1)} = \mathbf{x}_i^{(l,\text{attn})} + F^{(l)}(\mathbf{x}_i^{(l,\text{attn})}).$$

Let $\mathbf{x}_i^{(l,\text{attn,RSP})}$ and $\mathbf{x}_i^{(l,\text{attn,base})}$ denote the post-attention states under RSP and the unperturbed forward pass at the same input $\mathbf{x}_i^{(l)}$.

Lemma 4 (Per-block FFN propagation). *The post-attention difference satisfies*

$$\|\mathbf{x}_i^{(l,\textit{attn,RSP})} - \mathbf{x}_i^{(l,\textit{attn,base})}\| \le \|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)}\|\tilde{\mathbf{x}}_i^{(l+1)}\|.$$

If, in addition, $F^{(l)}$ is $L^{(l)}$-Lipschitz on a bounded set containing both $\mathbf{x}_i^{(l,\textit{attn,RSP})}$ and $\mathbf{x}_i^{(l,\textit{attn,base})}$, then

$$\|\mathbf{x}_i^{(l+1,\textit{RSP})} - \mathbf{x}_i^{(l+1,\textit{base})}\| \le (1+L^{(l)})\big(\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)}\|\tilde{\mathbf{x}}_i^{(l+1)}\|\big).$$

Proof. The unperturbed attention sublayer produces $\mathbf{x}_i^{(l,\text{attn,base})} = \mathbf{x}_i^{(l)} + \tilde{\mathbf{x}}_i^{(l+1)}$, while the RSP attention sublayer produces $\mathbf{x}_i^{(l,\text{attn,RSP})} = \mathbf{x}_i^{(l)} + (1-w_{r,i}^{(l)})\tilde{\mathbf{x}}_i^{(l+1)} + \boldsymbol{\eta}_i^{(l)}$ by Proposition 6. Subtracting,

$$\mathbf{x}_i^{(l,\text{attn,RSP})} - \mathbf{x}_i^{(l,\text{attn,base})} = \boldsymbol{\eta}_i^{(l)} - w_{r,i}^{(l)}\tilde{\mathbf{x}}_i^{(l+1)},$$

and the triangle inequality gives the first bound. For the second bound, write the post-FFN difference as

$$\big[\mathbf{x}_i^{(l,\text{attn,RSP})} + F^{(l)}(\mathbf{x}_i^{(l,\text{attn,RSP})})\big] - \big[\mathbf{x}_i^{(l,\text{attn,base})} + F^{(l)}(\mathbf{x}_i^{(l,\text{attn,base})})\big].$$

By Lipschitz continuity, $\|F^{(l)}(\mathbf{x}^{(l,\text{attn,RSP})}) - F^{(l)}(\mathbf{x}^{(l,\text{attn,base})})\| \le L^{(l)}\|\mathbf{x}_i^{(l,\text{attn,RSP})} - \mathbf{x}_i^{(l,\text{attn,base})}\|$. The triangle inequality combined with the first bound gives the second. ☐

Corollary 7 (Cross-block propagation control). *Define $\Delta^{(l)} := \mathbf{x}_i^{(l,\textit{RSP})} - \mathbf{x}_i^{(l,\textit{base})}$ with $\Delta^{(0)} = \mathbf{0}$ (matched inputs at layer 0), and let $L_{\textit{block}}^{(l)}$ denote a common local Lipschitz constant valid for both the baseline block map $B_{\textit{base}}^{(l)} : \mathbf{x} \mapsto \mathbf{x} + \mathrm{Attn}^{(l)}(\mathbf{x}) + F^{(l)}(\mathbf{x} + \mathrm{Attn}^{(l)}(\mathbf{x}))$ and the RSP-augmented block map $B_{\textit{RSP}}^{(l)}$ (which replaces $\mathrm{Attn}^{(l)}$ with the attention computation that includes the random tokens) on a ball containing both trajectories. Such a constant exists for standard transformer blocks because Attn, LayerNorm, and the FFN sublayer are each locally Lipschitz on any bounded set [Xiong et al., 2020, Kim et al., 2021], and the random-key augmentation only adds a finite number of additional softmax inputs at every layer. Then*

$$\|\Delta^{(l+1)}\| \le (1+L_{\textit{block}}^{(l)})\,\|\Delta^{(l)}\| + (1+L^{(l)})\big(\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)}\|\tilde{\mathbf{x}}_i^{(l+1)}\|\big).$$

Iterating across L blocks with $\Delta^{(0)} = \mathbf{0}$ yields

$$\|\Delta^{(L+1)}\| \le \sum_{l=0}^{L}\prod_{l'=l+1}^{L}(1+L_{\textit{block}}^{(l')})\cdot(1+L^{(l)})\big(\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)}\|\tilde{\mathbf{x}}_i^{(l+1)}\|\big).$$

Proof. Decompose the cross-block discrepancy at layer $l+1$ into two contributions:

$$\Delta^{(l+1)} = \underbrace{B_{\text{RSP}}^{(l)}(\mathbf{x}^{(l,\text{RSP})}) - B_{\text{RSP}}^{(l)}(\mathbf{x}^{(l,\text{base})})}_{\text{(I) propagation of prior }\Delta^{(l)}} + \underbrace{B_{\text{RSP}}^{(l)}(\mathbf{x}^{(l,\text{base})}) - B_{\text{base}}^{(l)}(\mathbf{x}^{(l,\text{base})})}_{\text{(II) RSP-induced perturbation injected at block }l},$$

where $B_{\text{RSP}}^{(l)}$ replaces $\mathrm{Attn}^{(l)}$ with the RSP-augmented attention. Term (I) is bounded via the block Lipschitz constant: $\|(I)\| \le L_{\text{block}}^{(l)}\|\Delta^{(l)}\|$. With residuals included one gets the factor $(1+L_{\text{block}}^{(l)})$ when also propagating the residual term $\Delta^{(l)}$ itself. Term (II) is exactly the per-block injection from Lemma 4, bounded by $(1+L^{(l)})\big(\|\boldsymbol{\eta}_i^{(l)}\| + w_{r,i}^{(l)}\|\tilde{\mathbf{x}}_i^{(l+1)}\|\big)$. Combining via the triangle inequality gives the recursion. The summed bound follows by unrolling the recursion: with $\Delta^{(0)} = \mathbf{0}$, induction on L shows that $\|\Delta^{(L+1)}\|$ is at most the displayed sum. ☐

Boundedness of $L^{(l)}$ for standard architectures. Modern transformer blocks combine LayerNorm, GELU or SwiGLU activations, and linear projections. LayerNorm has Lipschitz constant on the order of $\|\gamma\|/\inf\|x\|$ on every set bounded away from the origin, where γ is the LayerNorm scale parameter [Xiong et al., 2020]. GELU and SwiGLU are smooth and therefore locally Lipschitz on bounded activation regions, and linear projections have Lipschitz constant equal to the spectral norm of the weight matrix. The composed self-attention sublayer has a Lipschitz constant that depends on the spectral norms of $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ and the inverse-temperature β as analyzed by Kim et al. [2021]. On the bounded activation regions traversed during a single forward pass, the per-block Lipschitz constants in Corollary 7 are therefore finite. For a fixed model depth this yields a finite cumulative bound on $\|\Delta^{(L+1)}\|$; stronger depth-uniform non-expansion would require additional contraction assumptions not imposed here. This is the formal sense in which the injected perturbation stays controlled across a finite forward pass, while the size of each new injection is set by $w_{r,i}^{(l)}$.

E.4 Proofs for Output Distribution Shift

The main text makes two output-level claims. Proposition 1 establishes pairwise rank randomness, and Proposition 2 upgrades that statement to top-K entry. Both are statements about a local first-order surrogate of the perturbed logits.

E.4.1 Autoregressive Formulation

In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^{T} p(y_t \mid x, y_{<t}),$$

where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single global distribution.

E.4.2 Local Linearization of Logits under RSP

Throughout this subsection, $\bar{h} \in \mathbb{R}^d$ denotes *one* centered prompt vector $\bar{h}_l$ from Definition 1, treated as a per-token surrogate. The actual implementation uses a length-L prompt $\mathbf{H} = (h_1, \ldots, h_L)$ with L independent draws; the propositions below identify how each individual $\bar{h}_l$ shifts the logits, and the multi-token compounding through softmax-weighted attention is reported empirically in Figure 3 and Section 3.4. We model the perturbed logits as a function $z_i(\bar{h})$, assuming local smoothness in a neighborhood of $\bar{h} = 0$:

$$z_i(\bar{h}) = z_i(0) + \nabla_h z_i(0)^\top \bar{h} + r_i(\bar{h}),$$

where $r_i(\bar{h}) = O(\|\bar{h}\|^2)$. Defining $a_i := \nabla_h z_i(0)$, we obtain the local affine surrogate

$$z_i^{\mathrm{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}.$$

This approximation is valid for sufficiently small $\|\bar{h}\|$ and is used as a first-order surrogate for the direction of perturbation rather than as an exact global model of the transformer logits.

For the two propositions below we isolate the only support property that is actually used. Let D denote the centered prompt law. We assume

$$\text{(S1)} \qquad D(U) > 0 \quad \text{for every nonempty open set } U \subseteq \mathbb{R}^d.$$

The Gaussian instantiation in Section 4.2 satisfies (S1), but the proofs below require only this support condition.

E.4.3 Proof of Proposition 1 (Stochastic Ranking)

If there exists a token pair (i, j) such that $a_i \neq a_j$, and if the centered prompt law satisfies (S1), then $0 < \Pr_{\bar{h}}(z_i^{\mathrm{lin}}(\bar{h}) > z_j^{\mathrm{lin}}(\bar{h})) < 1$.

Proof. Under the local affine surrogate,

$$z_i^{\mathrm{lin}}(\bar{h}) - z_j^{\mathrm{lin}}(\bar{h}) = (z_i - z_j) + (a_i - a_j)^\top \bar{h}.$$

Let $b_{ij} := a_i - a_j \neq 0$. The perturbed logit difference

$$\Delta_{ij}(\bar{h}) := (z_i - z_j) + b_{ij}^\top \bar{h}$$

vanishes on the affine hyperplane

$$H_0 := \{\bar{h} \in \mathbb{R}^d : (z_i - z_j) + b_{ij}^\top \bar{h} = 0\}.$$

Because $b_{ij} \neq 0$, H_0 has codimension one and partitions $\mathbb{R}^d$ into the two nonempty open half-spaces

$$H_+ := \{\bar{h} : (z_i - z_j) + b_{ij}^\top \bar{h} > 0\}, \qquad H_- := \{\bar{h} : (z_i - z_j) + b_{ij}^\top \bar{h} < 0\}.$$

By (S1), both H_+ and H_- have positive probability, which yields

$$0 < \Pr_{\bar{h}}(\Delta_{ij}(\bar{h}) > 0) < 1.$$

□

E.4.4 Proof of Proposition 2 (Top-K Entry)

Let $J_K(s_t)$ be the baseline top-K set at state s_t, and let $i \notin J_K(s_t)$. Define

$$\mathcal{G}_{i,K} := \left\{ \bar{h} \in \mathbb{R}^d : \# \left\{ j \neq i : z_j^{\mathrm{lin}}(\bar{h}) > z_i^{\mathrm{lin}}(\bar{h}) \right\} \leq K - 1 \right\}.$$

If $\mathcal{G}_{i,K}$ contains a nonempty open set and the centered prompt law satisfies (S1), then

$$\Pr_{\bar{h}} \left(\bar{h} \in \mathcal{G}_{i,K} \right) > 0.$$

Proof. Under the local affine surrogate, token i belongs to the surrogate top-K set if and only if at most $K-1$ tokens outrank it, i.e., if and only if $\bar{h} \in \mathcal{G}_{i,K}$. For each subset S of token indices with $i \notin S$ and $|S| \leq K-1$, define

$$\mathcal{O}_S := \left\{ \bar{h} : z_j^{\mathrm{lin}}(\bar{h}) > z_i^{\mathrm{lin}}(\bar{h}) \text{ for all } j \in S,\ z_i^{\mathrm{lin}}(\bar{h}) > z_j^{\mathrm{lin}}(\bar{h}) \text{ for all } j \notin S \cup \{i\} \right\}.$$

Each $\mathcal{O}_S$ is an intersection of open half-spaces and is therefore an open subset of $\mathbb{R}^d$. By construction, every $\bar{h} \in \mathcal{O}_S$ has token i ranked $|S| + 1 \leq K$ in the affine surrogate, so $\mathcal{O}_S \subseteq \mathcal{G}_{i,K}$. Hence

$$\bigcup_{S:\, i \notin S,\ |S| \leq K-1} \mathcal{O}_S \ \subseteq\ \mathcal{G}_{i,K}.$$

The reverse inclusion fails only on the measure-zero set of $\bar{h}$ that produce ties, which we do not need: the assumption that $\mathcal{G}_{i,K}$ contains a nonempty open set is sufficient. Condition (S1) therefore implies

$$\Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0,$$

which is exactly the event that token i enters the surrogate top-K set. □

E.4.5 Proof of Corollary 2 (Sample Complexity)

Let $\mu_i = \Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0$. Within N independent RSP samples $\bar{h}_1, \ldots, \bar{h}_N$, the probability of at least one admission is at least $1 - (1-\mu_i)^N \geq 1 - \exp(-N\mu_i)$.

Proof. The samples $\{\bar{h}_n\}_{n=1}^N$ are independent, so

$$\Pr\big(\text{no admission within } N \text{ samples}\big) = \prod_{n=1}^{N} \Pr(\bar{h}_n \notin \mathcal{G}_{i,K}) = (1-\mu_i)^N.$$

The complement gives the first inequality. The second inequality is standard: $1 - \mu_i \leq \exp(-\mu_i)$, hence $(1-\mu_i)^N \leq \exp(-N\mu_i)$. □

E.4.6 Proof of Proposition 3 (Exploration-meets-multi-path Pass@N gain)

Fix a problem x and let $p_n(x) \geq p_{\min}(x) > 0$ be the per-run success probability of the n-th RSP run as in (M1). Under (M1)+(M2),

$$\Pr\big(\text{Pass@}(N{+}1) \text{ correct on } x\big) \ \geq\ \max\big\{p_{\mathrm{base}}(x),\ 1 - (1-p_{\min}(x))^N\big\} \ \geq\ 1 - e^{-Np_{\min}(x)}.$$

Proof. Let A_0 be the event that the deterministic baseline run reaches a trajectory in $\mathcal{T}^\star(x)$, and let A_n ($n = 1, \ldots, N$) be the event that the n-th RSP run reaches such a trajectory. Pass@$(N+1)$ succeeds on x iff at least one of $A_0, A_1, \ldots, A_N$ occurs.

By assumption $p_{\mathrm{base}}(x) := \Pr(A_0) \in \{0,1\}$ and $\Pr(A_n) = p_n(x) \geq p_{\min}(x)$ for each $n \geq 1$. Independence of the N RSP runs (M2) gives

$$\Pr\Big(\bigcap_{n=1}^{N} A_n^c\Big) = \prod_{n=1}^{N}(1 - p_n(x)) \leq \prod_{n=1}^{N}(1 - p_{\min}(x)) = (1 - p_{\min}(x))^N,$$

using $p_n(x) \geq p_{\min}(x)$ in the inequality. Therefore

$$\Pr\Big(\bigcup_{n=1}^{N} A_n\Big) = 1 - \Pr\Big(\bigcap_{n=1}^{N} A_n^c\Big) \geq 1 - (1 - p_{\min}(x))^N.$$

The Pass@$(N+1)$ event $A_0 \cup A_1 \cup \cdots \cup A_N$ has probability at least $\max\{\Pr(A_0),\, \Pr(\bigcup_{n\geq 1} A_n)\}$, giving the first claimed inequality. The second inequality follows from the standard bound $1-x \leq e^{-x}$ for $x \in [0,1]$ applied with $x = p_{\min}(x)$:

$$(1 - p_{\min}(x))^N \leq e^{-N p_{\min}(x)},$$

hence $1 - (1 - p_{\min}(x))^N \geq 1 - e^{-N p_{\min}(x)}$. $\square$

E.4.7 Trajectory-Level Compounding

Since autoregressive generation composes conditional distributions across steps, perturbations at early steps can lead to distinct trajectories. Let $\mathcal{R}_{k,K}(x)$ denote the set of trajectories containing at least one token outside the baseline top-K set. If RSP increases the accessibility of such tokens at early steps, then the probability of sampling trajectories in $\mathcal{R}_{k,K}(x)$ increases under repeated sampling. The key point is temporal compounding: one early top-K entry changes the next decoding state, which changes the next candidate set, and so on. This is the heuristic bridge from local rank perturbations to global trajectory diversity that motivates assumption (M1) of Proposition 3.

F Attention Mass Analysis

This appendix formalizes the per-token attention mass reported in Figure 3 and the exclusion criteria applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both problem variability and RSP-vector variability.

F.1 Per-Token Attention Mass

For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,h)}$ denote the post-softmax attention weight at layer ℓ, head h, from query position i to key position j, satisfying $\sum_j \alpha_{i,j}^{(\ell,h)} = 1$ under the causal mask. Let Q and R denote the index sets of question and RSP tokens with sizes $|Q|$ and $|R| = L$, and let H be the number of heads. The per-token attention mass that query i directs to each group at layer ℓ is

$$\mathrm{PTM}_Q[\ell, i] = \frac{1}{|Q|H} \sum_{h=1}^{H} \sum_{j \in Q} \alpha_{i,j}^{(\ell,h)}, \qquad \mathrm{PTM}_R[\ell, i] = \frac{1}{|R|H} \sum_{h=1}^{H} \sum_{j \in R} \alpha_{i,j}^{(\ell,h)}. \tag{9}$$

Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a single question (resp. RSP) token receives on average under the same softmax denominator. Question tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both groups.

F.2 Heatmap Aggregation

We partition the layer indices and the reasoning position indices each into five equal-size bins $\{B_L^{(b)}\}_{b=1}^5$ and $\{B_P^{(b)}\}_{b=1}^5$. For sample s and target group $\bullet \in \{Q, R\}$, the value in cell (b_L, b_P) is

$$M_\bullet^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \mathrm{PTM}_\bullet[\ell, i]. \tag{10}$$

Figure 3 reports the sample average of Eq. (10) over the 500 valid samples.

F.3 Excluding the `\boxed{...}` Span

The reasoning position range terminates strictly before the final `\boxed{...}` span. Chain-of-thought outputs decompose into a *text* (content) component and a *pattern* (template) component that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The `\boxed{...}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the answer format rather than to extend reasoning. Including it would therefore inject a template-driven attention signature unrelated to reasoning dynamics.

F.4 Excluding the Final Two Layers

The layer axis further excludes the last two transformer layers. Two well-documented properties of late layers, none specific to our setup, motivate this choice.

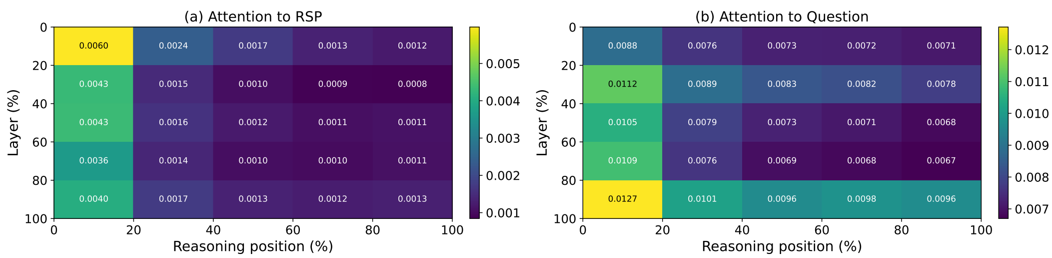
Output-projection specialization. Late-layer hidden states lie in an approximately linear relationship with vocabulary logits and therefore function as a progressive linear readout rather than as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021]. Attention in these layers therefore reflects output formation rather than reasoning computation.

Register-token hypothesis. Uninformative patch tokens in Vision Transformers have been shown to be repurposed as *registers*: storage slots that absorb attention in deeper layers [Darcet et al., 2024]. Because RSPs share a similar profile as continuous tokens without pretrained semantic content, a similar repurposing in late transformer layers cannot be ruled out. Excluding the final two layers removes this potential register-induced confound from the reasoning-phase signal.

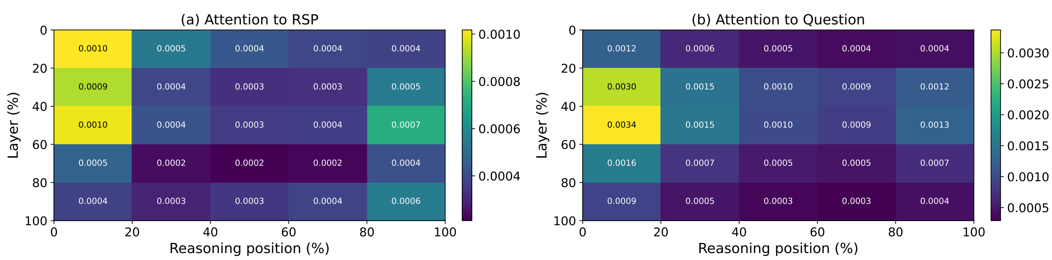
The two lines of evidence above identify known late-layer phenomena — linear readout and potential register repurposing — whose attention signatures are orthogonal to reasoning-phase dynamics. Removing the final two layers preempts their contamination of the reasoning-phase signal, yielding a principled, literature-grounded exclusion rather than a model-specific tuning decision.

F.5 Additional Suffix Heatmaps

Figure 5 reports the same per-token RSP attention analysis under suffix injection for Qwen2.5-Math-1.5B-Instruct and LLaMA-3.1-8B-Instruct. For the late-layer reasons discussed above, the pattern does not generalize uniformly across every cell; nevertheless, the overall trend is consistent across both models: question-token attention varies broadly across layers, whereas RSP attention shows the layer-wise and sequence-wise decay predicted by Eq. (4).



(a) Qwen2.5-Math-1.5B-Instruct, suffix



(b) LLaMA-3.1-8B-Instruct, suffix

Figure 5: Per-token RSP attention mass under suffix injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 3: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

G GRPO Formulation

For a given prompt q, the behavior policy $\pi_{\theta_{\text{old}}}$ samples a group of G responses $\{o_i\}_{i=1}^{G}$ with corresponding rewards $\{R_i\}_{i=1}^{G}$. The advantage of each response is computed by normalizing the group-level rewards:

$$\hat{A}_{i,t} = \frac{R_i - \text{mean}(\{R_i\}_{i=1}^{G})}{\text{std}(\{R_i\}_{i=1}^{G})}. \tag{11}$$

The policy is updated via a clipped objective with a KL penalty:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\Big(\min\Big(r_{i,t}\hat{A}_{i,t},\ \text{clip}(r_{i,t}, 1-\varepsilon, 1+\varepsilon)\,\hat{A}_{i,t}\Big) - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}})\Big)\right], \tag{12}$$

where $r_{i,t} = \pi_\theta(o_{i,t} \mid q, o_{i,<t})/\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})$ is the importance sampling ratio, ε is the clipping range, and β controls the strength of the KL penalty against the reference policy π_{ref}.

H Broader Impacts

This work provides empirical and theoretical analysis of how random embedding injection affects the reasoning behavior of large language models. The contributions are primarily foundational, with potential downstream implications for reinforcement learning-based post-training of reasoning models such as GRPO.

Positive impacts. RSP-induced trajectory diversity could improve the sample efficiency of RL-based LLM post-training by providing richer learning signals within group-relative advantage estimation. This has the potential to reduce compute costs associated with alignment and reasoning-oriented fine-tuning, making such methods more accessible to research groups with limited resources.

Potential concerns. Methods that expand the effective output support of LLMs may increase the reachability of low-probability tokens, which includes both desirable alternative reasoning paths and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems should combine it with existing safety filtering and alignment mechanisms rather than treating it as a standalone diversity source.

Limitations on impact scope. Since this work focuses on analysis rather than deploying a new model or dataset, the direct societal impact is limited. The broader implications described above are contingent on future work integrating RSP into applied training pipelines.

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